

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

FORMAL DIGITAL FORENSICS INVESTIGATION REPORT

SBT-DF204 - Case Study 1: Investigating Harassment Email Traffic with Wireshark

Field	Detail
Case Identifier	SBT-DF204-CaseStudy1
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Registration / ID No.	202520850LE
Course Code	SBT-DF204
Course Title	Computer Forensics Case Studies
Case Title	Investigating Harassment Email Traffic With Wireshark
Date of Submission	October 5, 2026
Primary Analysis System	Kali Linux virtual machine (VMware); Wireshark 4.6.6 and TShark 4.6.6
Evidence Type	Training packet capture nitroba.pcap (Nitroba University Harassment Scenario, Digital Corpora)

Prepared for authorized academic forensic examination. The capture was treated as controlled training material; no host, account or person referenced in it was contacted.

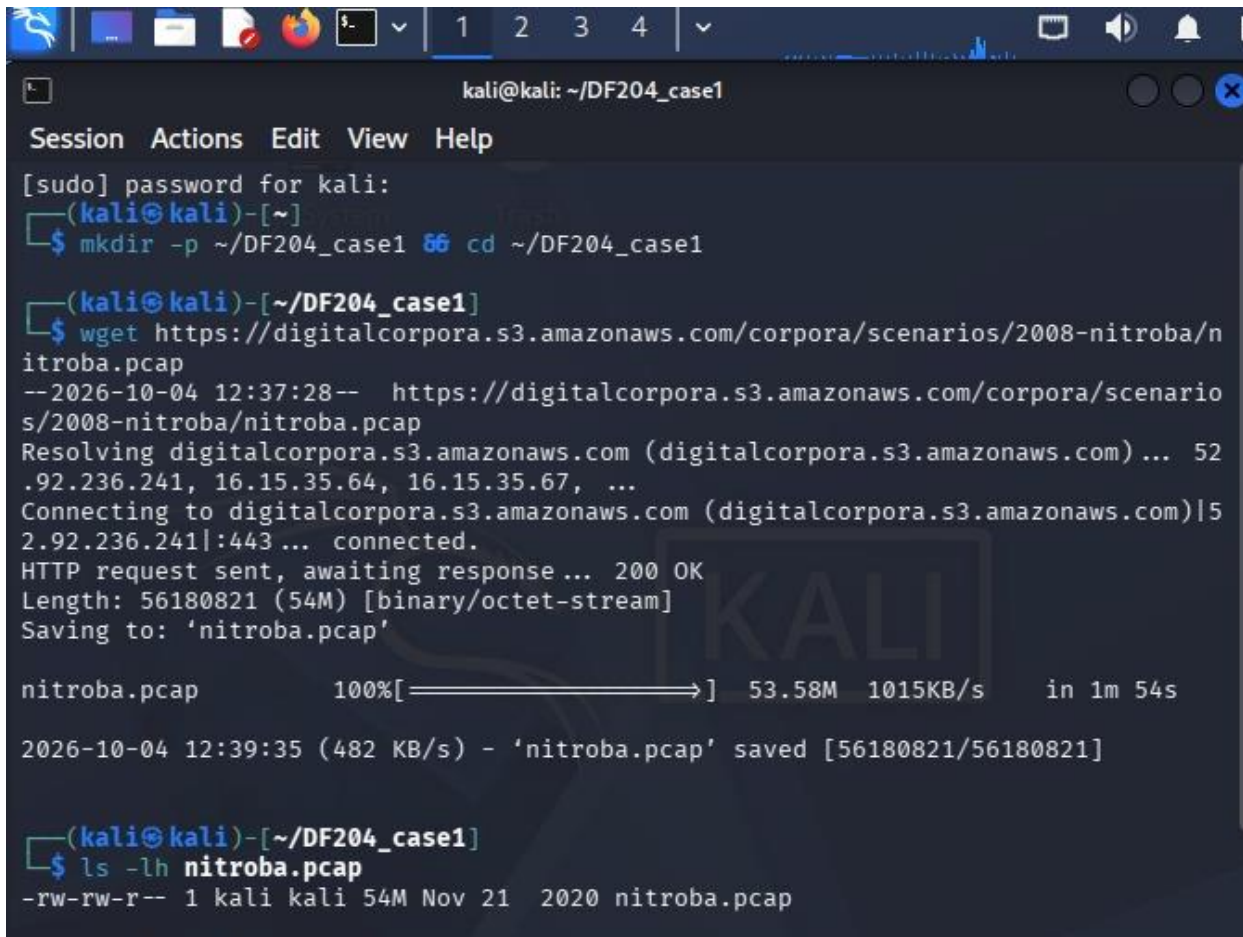
1. Executive Conclusion

The network evidence shows that client 192.168.15.4 contacted www.willselfdestruct.com and, from that client, an HTTP POST containing the harassment message was sent to 69.25.94.22. The same traffic is associated with Ethernet source MAC 00:17:f2:e2:c0:ce. A Gmail cookie containing jcoachj@gmail.com was also observed from that MAC in 109 HTTP packets, including shortly before and after the POST. These findings support attribution of the activity to the device and its network session. The association with a person is an inference: jcoachj@gmail.com resembles the Chem 109 roster name Johnny Coach, but the roster contains names only and does not independently verify ownership or use of the account. Overall confidence is moderate because the network, HTTP, MAC, and cookie evidence are mutually consistent, while the open shared wireless network prevents conclusive person-level identification.

2. Evidence Acquisition and Integrity

Item	Recorded value
Original file name	nitroba.pcap
Source link	https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap (scenario page: digitalcorpora.org/corpora/scenarios/nitroba-university-harassment-scenario)
File size	56,180,821 bytes (about 54 MB)
Download date and time	2026-10-04; wget finished 12:39:35 EDT (VM local time, UTC-4) = 16:39:35 UTC
File date shown by ls	Nov 21 2020 (last-modified time preserved by wget; not the download date)
SHA-256, original	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA-256, working copy	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA-1, working copy	65656392412add15f93f8585197a8998aaeb50a1 (shown by Wireshark)
Working copy method	cp nitroba.pcap nitroba_working.pcap; original made read-only with chmod 444 (ls -l shows -r--r--r--); cmp reported IDENTICAL; Wireshark Capture File Properties reports the same SHA-256
Published checksum	Published SHA-256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb. It matches the SHA-256 calculated for the original and working copy.
Capture properties	pcap format, Ethernet encapsulation, snapshot length 4096; first packet 2008-07-22 01:51:07 UTC; last packet 2008-07-22 06:13:47 UTC; elapsed 04:22:39

Time-zone note: the Capture File Properties window displayed the times in the VM local zone (EDT, UTC-4), e.g. first packet 2008-07-21 21:51:07 and last packet 2008-07-22 02:13:47. Four hours were added to give the UTC values above. The packet list was switched to View > Time Display Format > UTC Date and Time of Day, and all packet times in this report are taken from that view (shown with a trailing Z).



```

kali@kali: ~/DF204_case1
Session Actions Edit View Help
[sudo] password for kali:
(kali@kali)-[~]
$ mkdir -p ~/DF204_case1 56 cd ~/DF204_case1

(kali@kali)-[~/DF204_case1]
$ wget https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
--2026-10-04 12:37:28-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com) ... 52.92.236.241, 16.15.35.64, 16.15.35.67, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.236.241|:443... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'nitroba.pcap'

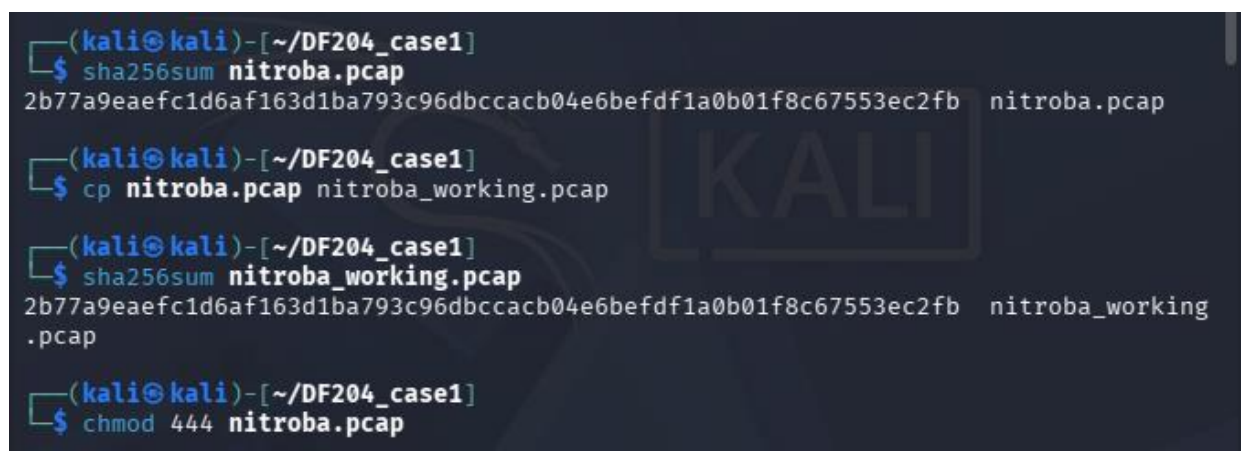
nitroba.pcap      100%[=====>] 53.58M  1015KB/s   in 1m 54s

2026-10-04 12:39:35 (482 KB/s) - 'nitroba.pcap' saved [56180821/56180821]

(kali@kali)-[~/DF204_case1]
$ ls -lh nitroba.pcap
-rw-rw-r-- 1 kali kali 54M Nov 21  2020 nitroba.pcap

```

Fig01a. Download of nitroba.pcap with wget (56,180,821 bytes) and file listing.



```

(kali@kali)-[~/DF204_case1]
$ sha256sum nitroba.pcap
2b77a9eafc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb nitroba.pcap

(kali@kali)-[~/DF204_case1]
$ cp nitroba.pcap nitroba_working.pcap

(kali@kali)-[~/DF204_case1]
$ sha256sum nitroba_working.pcap
2b77a9eafc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb nitroba_working.pcap

(kali@kali)-[~/DF204_case1]
$ chmod 444 nitroba.pcap

```

Fig01b. SHA-256 of the original, creation of the working copy, SHA-256 of the copy, and chmod 444 on the original.

```

kali@kali: ~/DF204_case1
Session Actions Edit View Help

(kali@kali)-[~/DF204_case1]
$ cmp nitroba.pcap nitroba_working.pcap 86 echo "IDENTICAL"
IDENTICAL

(kali@kali)-[~/DF204_case1]
$ ls -l nitroba.pcap
-r--r--r-- 1 kali kali 56180821 Nov 21 2020 nitroba.pcap

(kali@kali)-[~/DF204_case1]
$ date; date -u
Sun Oct 4 05:20:47 PM EDT 2026
Sun Oct 4 09:20:47 PM UTC 2026

(kali@kali)-[~/DF204_case1]
$ wireshark nitroba_working.pcap &
[1] 19977

(kali@kali)-[~/DF204_case1]
$ ** (wireshark:19977) 17:21:34.280262 [GUI WARNING] -- Failed to register with host portal QDBusError("org.freedesktop.portal.Error.Failed", "Could not register app ID: Connection already associated with an application ID")

```

Fig01c. cmp reports IDENTICAL; the original is read-only (-r--r--r--); date shows EDT and UTC; Wireshark opened on the working copy.

Wireshark · Capture File Properties · nitroba_working.pcap

Details

File

- Name: nitroba_working.pcap
- Length: 56 MB
- Hash (SHA256): 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
- Hash (SHA1): 65656392412add15f93f8585197a8998aaeb50a1
- Format: Wireshark/tcpdump/... - pcap
- Encapsulation: Ethernet
- Snapshot length: 4096

Time

- First packet: 2008-07-21 21:51:07
- Last packet: 2008-07-22 02:13:47
- Elapsed: 04:22:39

Capture

- Hardware: Unknown
- OS: Unknown
- Application: Unknown

Interfaces

Interface	Interface Description	Dropped packets	Capture filter	Link type	Packet size limit (snaplen)

Fig02a. Statistics > Capture File Properties: SHA-256, SHA-1, pcap format, Ethernet encapsulation, first and last packet (VM local time) and elapsed time.

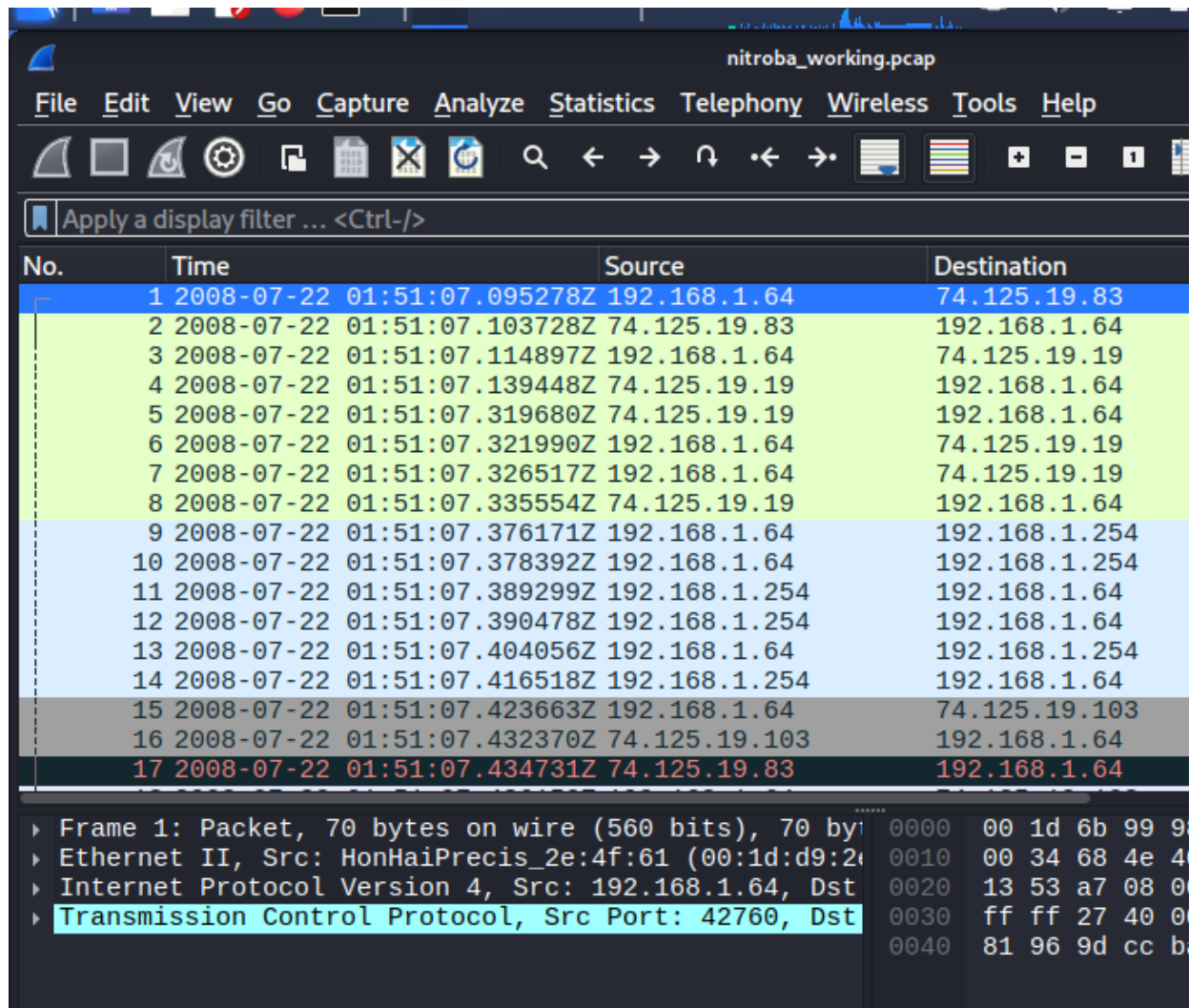


Fig02b. Packet list with the time display set to UTC (trailing Z); first packet 2008-07-22 01:51:07.095278Z.

3. Method

#	Filter / command	Result
1	sha256sum, cp, cmp, chmod 444	Hashes of original and copy identical; original read-only (E01)
2	Statistics > Capture File Properties	Same SHA-256 in Wireshark; capture 01:51:07 to 06:13:47 UTC (E02)
3	frame contains "willselfdestruct"	Wide search: many destinations; first hit frame 82897; includes DNS and HTTP (E03, E04)
4	ip.addr == 69.25.94.22 && http	GET frame 82936 and 200 OK frame 82998 (E04, E05)
5	ip.addr == 69.25.94.22 && http.request.method == "POST"	One packet: frame 83601 (E06)
6	Follow TCP Stream, tcp.stream eq 1707	Full POST request and 302 response (E07)

#	Filter / command	Result
7	eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"	Gmail cookie in frame 83326; Amazon and ad-network false positives (E10, E13)
8	TShark count of http.cookie contains "jcoachj"	109 packets, frames 77528 to 84201 (E11)
9	TShark email-pattern search excluding jcoachj	0 packets (E12)

The examination began with the complete packet capture and preservation of the original evidence through hashing, comparison with a working copy, and read-only protection of the original. I then searched the capture for references to willselfdestruct.com and followed the DNS resolution to identify the service IP address and the client that contacted it. I narrowed the traffic to the relevant HTTP GET and then to the HTTP POST containing the harassment message. I next correlated the client IP with the Ethernet source MAC address and examined HTTP cookies originating from that MAC. Search results containing characters such as "@" were treated as leads rather than findings because symbols can occur inside unrelated tracking tokens; the apparent matches were therefore inspected and compared with their context. The key results were cross-checked using both Wireshark and TShark to reduce the risk of relying on a single tool or filter.

4. Findings

Q1. What file did you acquire, and how did you preserve its integrity?

See Section 2 and evidence E01, E02 (Figures 1 and 2). Original and working copy share one SHA-256; the working copy is the only file analysed.

Q2. Which client system contacted the web service?

Item	Value	Supporting packets and filter
Client IP	192.168.15.4	Source of DNS query 82912 and GET 82936; filter frame contains "willselfdestruct"
Service IP	69.25.94.22	DNS answer (A record) in 82923; destination of GET 82936; source of 200 OK 82998; filter ip.addr == 69.25.94.22 && http
DNS resolver	192.168.1.254	Query 82912 and response 82923 (query for www.willselfdestruct.com, answered via CNAME to willselfdestruct.com, A 69.25.94.22)

nitroba_working.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

frame contains "willselfdestruct"

No.	Time	Source	Destination
82897	2008-07-22 06:03:43.528643Z	192.168.15.4	208.185.127.33
82898	2008-07-22 06:03:43.549020Z	208.185.127.33	192.168.15.4
82902	2008-07-22 06:03:43.551895Z	208.185.127.33	192.168.15.4
82906	2008-07-22 06:03:43.569272Z	192.168.15.4	208.185.127.40
82911	2008-07-22 06:03:43.615843Z	192.168.15.4	208.185.127.40
82912	2008-07-22 06:03:43.640466Z	192.168.15.4	192.168.1.254
82923	2008-07-22 06:03:43.720123Z	192.168.1.254	192.168.15.4
82925	2008-07-22 06:03:43.757607Z	192.168.15.4	208.185.127.35
82930	2008-07-22 06:03:43.798856Z	192.168.15.4	208.185.127.35
82936	2008-07-22 06:03:43.825871Z	192.168.15.4	69.25.94.22
82939	2008-07-22 06:03:43.848398Z	192.168.15.4	208.185.127.35
82940	2008-07-22 06:03:43.869763Z	192.168.15.4	64.236.76.160
82942	2008-07-22 06:03:43.870951Z	192.168.15.4	64.236.76.160
82944	2008-07-22 06:03:43.871756Z	192.168.15.4	192.217.199.107
82945	2008-07-22 06:03:43.873014Z	192.168.15.4	208.185.127.35
82947	2008-07-22 06:03:43.881289Z	192.168.15.4	208.185.127.35
82960	2008-07-22 06:03:43.947926Z	192.168.15.4	64.236.76.160

Frame 82897: Packet, 746 bytes on wire (5968 bits), 746 bytes captured (5968 bits) on interface 0

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: 208.185.127.33

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 208.185.127.33

Transmission Control Protocol, Src Port: 35962, Dst Port: 80

Hypertext Transfer Protocol

Fig03a. Filter frame contains "willselfdestruct": first matching packets from 192.168.15.4.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

frame contains "willselfdestruct"

No.	Time	Source	Destination	Protocol	Length	Info
82898	2008-07-22 06:03:43.549020Z	208.185.127.33	192.168.15.4	TCP	110	[TCP Previous segment not captured] 80 -> 35962 [ACK]
82902	2008-07-22 06:03:43.551895Z	208.185.127.33	192.168.15.4	TCP	1466	[TCP Fast Retransmission] 80 -> 35962 [ACK] Seq=1005
82906	2008-07-22 06:03:43.569272Z	192.168.15.4	208.185.127.40	HTTP	793	GET /g1/dynamic/offsite.htm?zi=1/XJ&sdn=email&cdn=com
82911	2008-07-22 06:03:43.615843Z	192.168.15.4	208.185.127.40	HTTP	1118	GET /g1/dynamic/offsetstetopad.htm?zi=1/XJ&sdn=email&cdn=com
82912	2008-07-22 06:03:43.640466Z	192.168.15.4	192.168.1.254	DNS	88	Standard query 0x0000 A www.willselfdestruct.com
82923	2008-07-22 06:03:43.720123Z	192.168.1.254	192.168.15.4	DNS	202	Standard query response 0x0000 A www.willselfdestruct.com
82925	2008-07-22 06:03:43.757607Z	192.168.15.4	208.185.127.35	HTTP	842	GET /6/js/b.txt?email HTTP/1.1
82930	2008-07-22 06:03:43.798856Z	192.168.15.4	208.185.127.35	HTTP	840	GET /6/g/email/b/b.js HTTP/1.1
82936	2008-07-22 06:03:43.825871Z	192.168.15.4	69.25.94.22	HTTP	596	GET /secure/submit HTTP/1.1
82939	2008-07-22 06:03:43.848398Z	192.168.15.4	208.185.127.35	HTTP	836	GET /f/lg/af1.gif HTTP/1.1
82940	2008-07-22 06:03:43.869763Z	192.168.15.4	64.236.76.160	HTTP	800	GET /rtx/r.js?cmd=ADN&si=11775&xs=2&v=3.12&cb=50904 HTTP/1.1
82942	2008-07-22 06:03:43.870951Z	192.168.15.4	64.236.76.160	HTTP	790	GET /dastat/ping.js?ADN&si=11775&cb=37601 HTTP/1.1
82944	2008-07-22 06:03:43.871756Z	192.168.15.4	192.217.199.107	TCP	1466	35912 -> 80 [ACK] Seq=2126 Ack=1185 Win=64768 Len=1406

Frame 82912: Packet, 1438 bytes on wire (11504 bits), 1438 bytes captured (11504 bits) on interface 0

Ethernet II, Src: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60), Dst: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 192.168.1.254

DNS Standard query 0x0000 A www.willselfdestruct.com

Type: IPv4 (0x0000)

[Stream index: 19]

Frame check sequence: 0x2c15ecf [unverified]

[FCS Status: Unverified]

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 192.168.1.254

Fig03b. Same filter with the Protocol and Info columns: DNS query 82912, DNS response 82923, HTTP GET /secure/submit 82936.

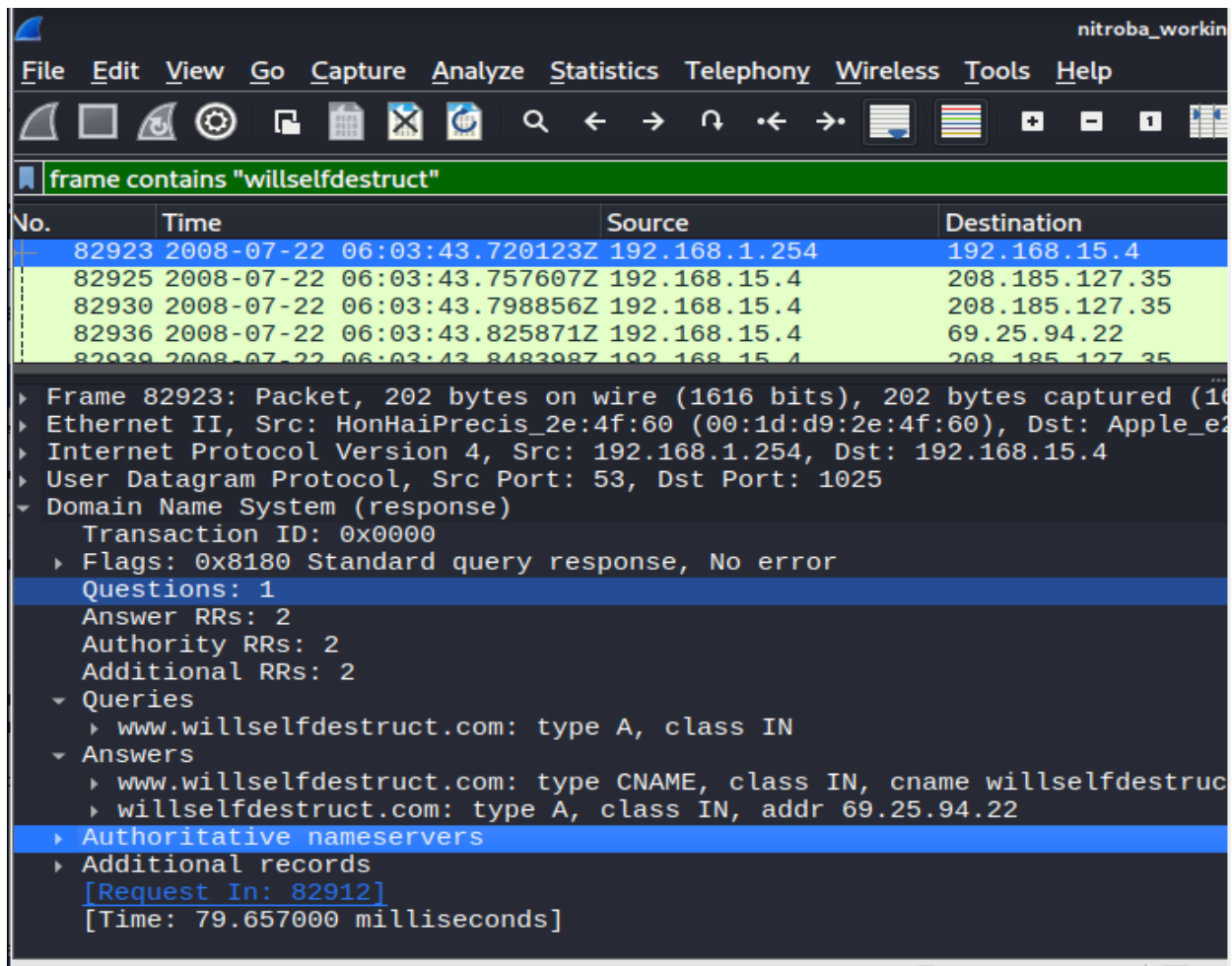


Fig04. DNS response, frame 82923: Answers show the CNAME and the A record 69.25.94.22.

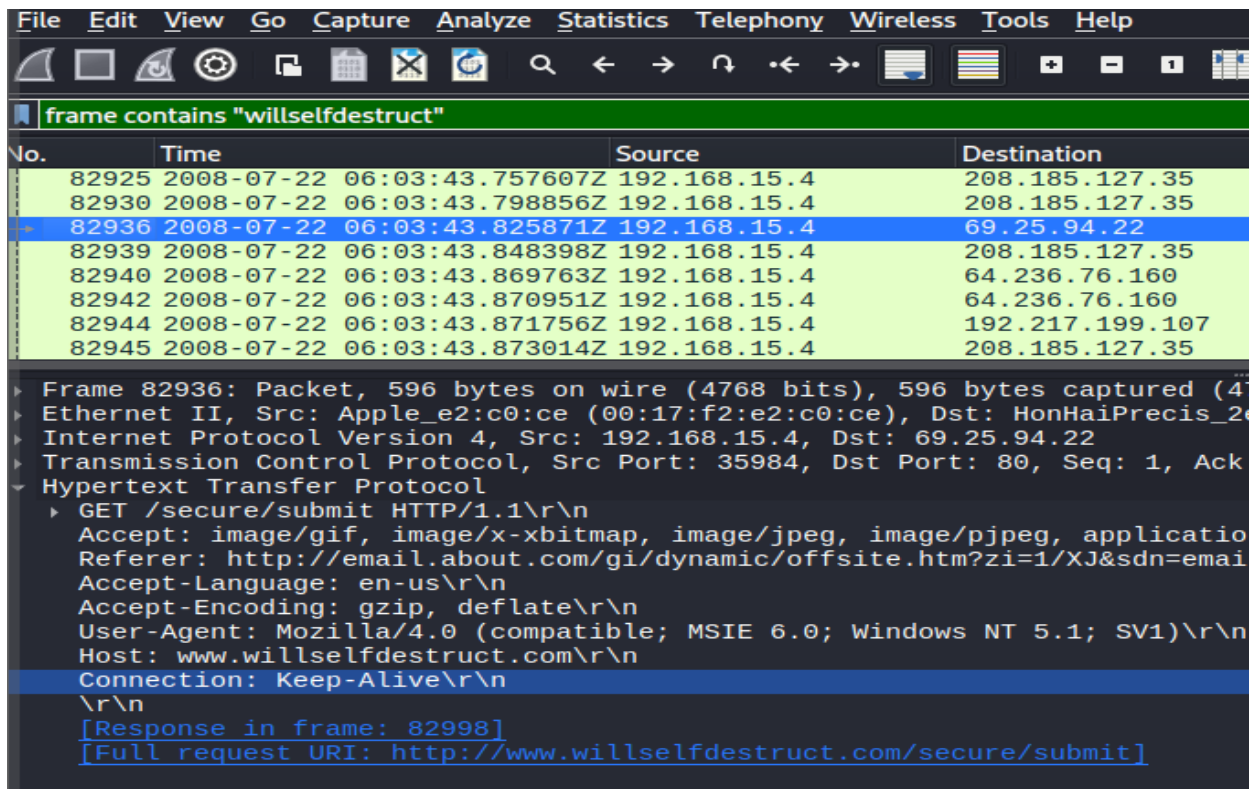


Fig05. HTTP GET, frame 82936: Host www.willselfdestruct.com and the Full request URI.

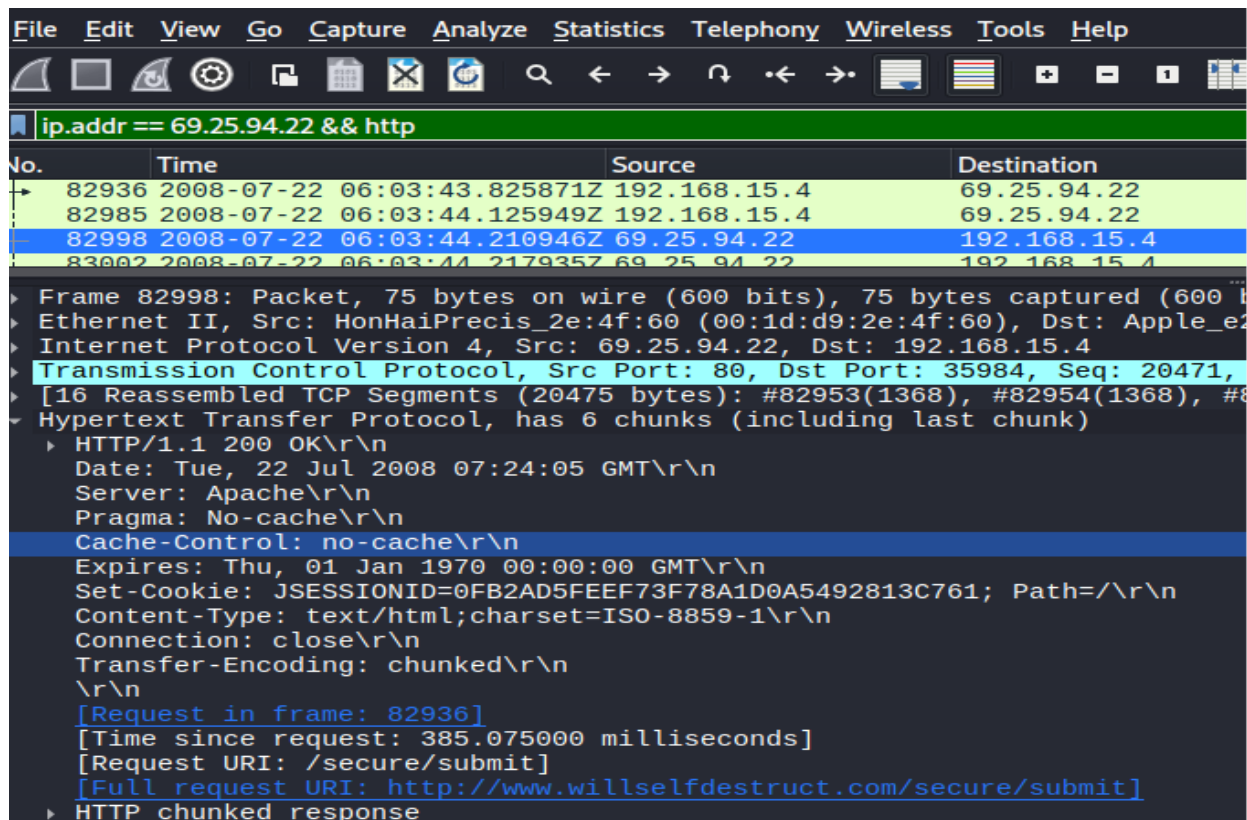


Fig06. HTTP/1.1 200 OK, frame 82998, filter ip.addr == 69.25.94.22 && http, with Request in frame 82936.

The DNS query and response, HTTP request, and HTTP response form a consistent sequence for the same web service. Frame 82912 shows the client querying for www.willselfdestruct.com, frame 82923 returns the CNAME and A record resolving the service to 69.25.94.22, and frame 82936 shows the client requesting the service at that IP address; frame 82998 then contains the corresponding HTTP response. This correlation supports the conclusion that 69.25.94.22 was the destination used for the [willselfdestruct.com](http://www.willselfdestruct.com) activity. The Host header in frame 82936 is important because it identifies the requested hostname, not merely the destination IP address; an IP address can host multiple services or domains, whereas the Host header provides application-level evidence of the specific website requested.

Q3. What evidence links the client to the harassment message?

Item	Observation
Packet / time	Frame 83601, 2008-07-22 06:04:24.311700 UTC, 192.168.15.4 to 69.25.94.22, 719 bytes, source port 36044
Request	POST /secure/submit HTTP/1.1, Host: www.willselfdestruct.com , Content-Length 188, stream tcp.stream eq 1707
Form fields	to = lilytuckrige@yahoo.com; from = (empty); subject = you can't find us; message = "and you can't hide from us." / "Stop teaching." / "Start running."; type = 0; ttl = 30; submit.x = 92; submit.y = 26

Item	Observation
Server response	HTTP/1.1 302 Moved Temporarily, Location: /secure/success. The response is visible in Follow TCP Stream (tcp.stream eq 1707); the supplied report did not independently record the response frame number, so no frame number has been invented.
Comparison	The To, Subject and three message lines match. The apostrophe in "can't" is consistent between the POST field and the teacher's displayed message. The sender is shown as [Undisclosed Sender], consistent with the empty from field.

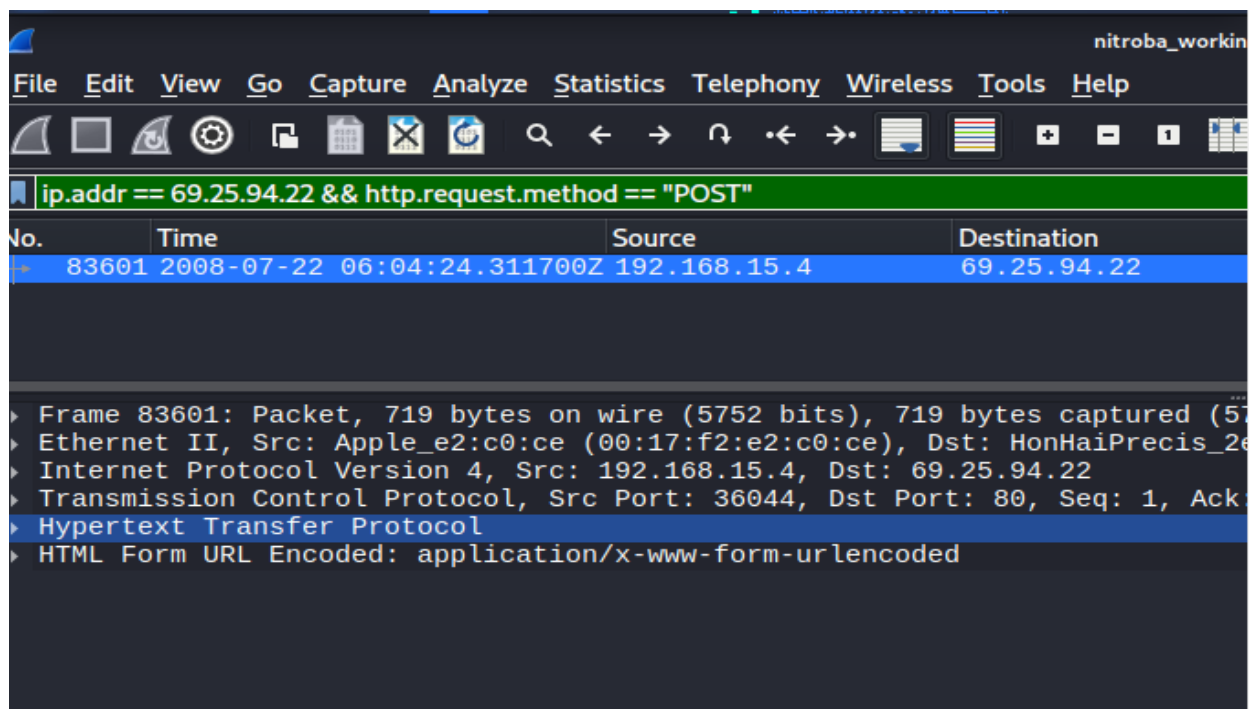
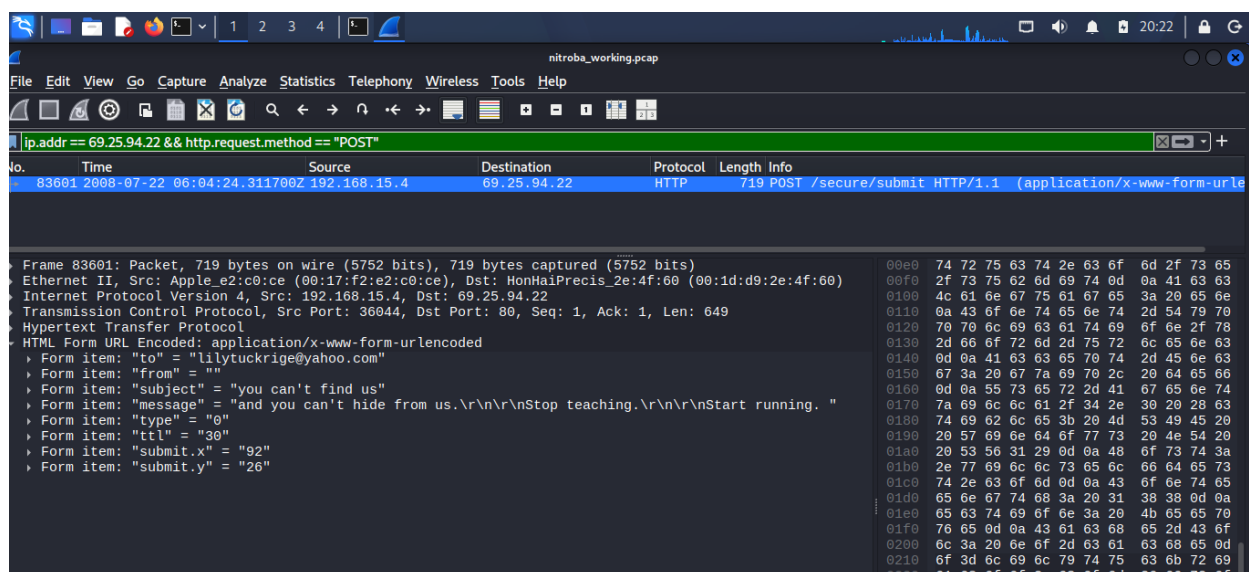
Fig07a. Filter `ip.addr == 69.25.94.22 && http.request.method == "POST"`: one packet, frame 83601.

Fig07b. Frame 83601, HTML Form URL Encoded: to, from, subject, message, type, ttl, submit.x and submit.y.

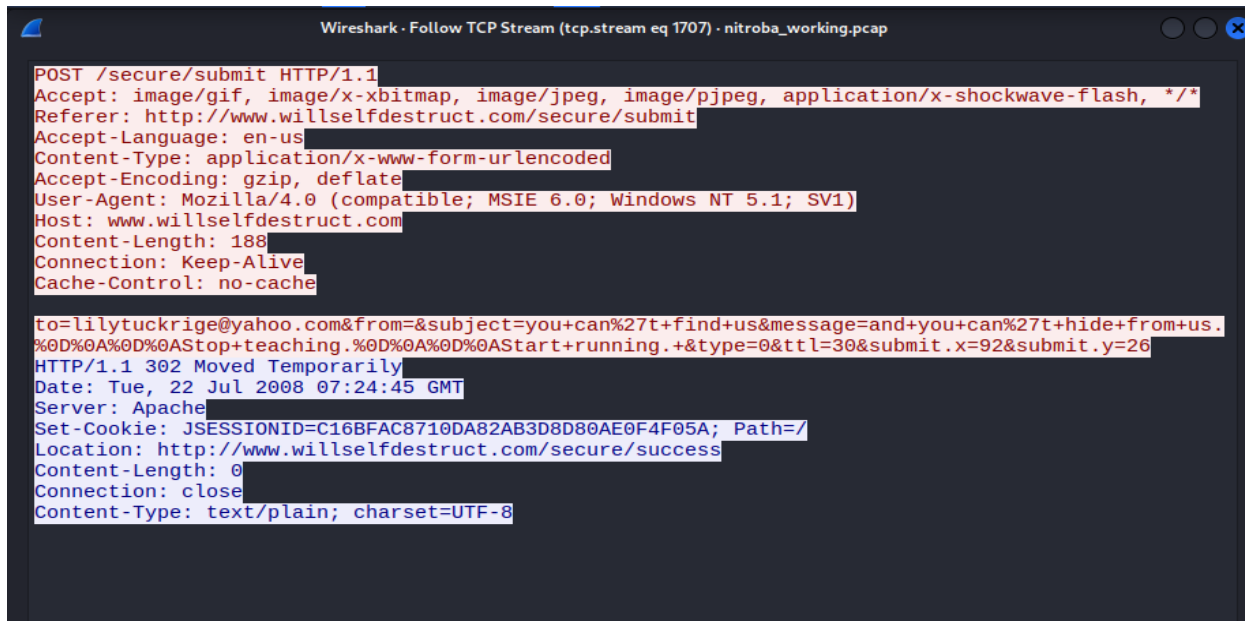


Fig08a. Follow TCP Stream (tcp.stream eq 1707): POST request and 302 Moved Temporarily response.

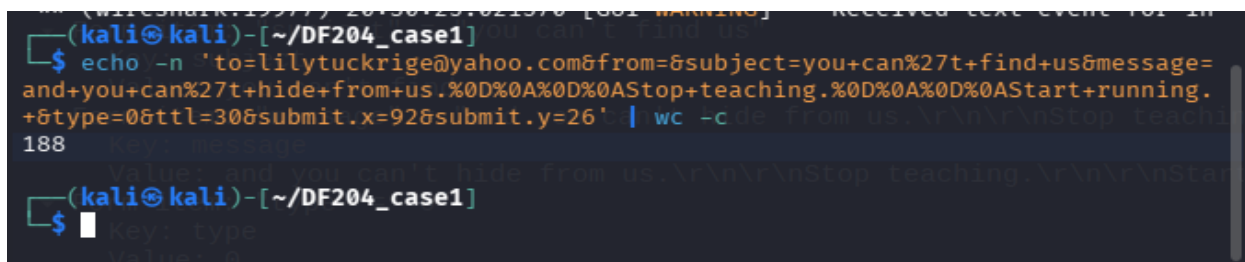


Fig08b. Body length check: the body is 188 bytes, matching Content-Length 188.



Fig09. The teacher's second email as displayed by willselfdestruct.com (From: [Undisclosed Sender]).

The POST request in frame 83601 contains the recipient, subject, and three message lines that match the teacher's second-email display in Figure 09. This establishes a strong content-level correlation between the captured web request and the message that was displayed to the recipient. It does not, by itself, prove the identity of the person who was operating the device, because the capture identifies network activity and a device rather than the human at the keyboard. The empty from field is also significant: the web service displayed the sender as "[Undisclosed Sender]", which is consistent with the POST containing no sender value. This explains why the message display does not independently identify the sender by name or email address.

Q4. Which device made the relevant request?

Item	Observation
Client MAC	00:17:f2:e2:c0:ce (Wireshark label Apple_e2:c0:ce), the Ethernet source of frames 82936, 83601, 83326 and 83476; filter eth.addr == 00:17:f2:e2:c0:ce
Next-hop MAC	00:1d:d9:2e:4f:60 (HonHaiPrecis_2e:4f:60), the Ethernet destination of the client frames and the source of DNS reply 82923
Browser identities from this MAC	MSIE 6.0 / Windows NT 5.1 in frames 82936, 83601, 83326; Firefox 2.0.0.16 / Mac OS X in frames 83476 and 59712; Safari 3.1.2 / Mac OS X 10_5_4 in frame 18823

No.	Time	Source	Destination	Protocol	Length	Info
83601	2008-07-22 06:04:24.311700Z	192.168.15.4	69.25.94.22	HTTP	719	POST /secure/submit HTTP/1.1 (application/x-www-form-url

Frame 83601: Packet, 719 bytes on wire (5752 bits), 719 bytes captured (5752 bits)	00e0	74	72	75	63	74	2e	63	6f	6d	2f	73	65
Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)	00f0	2f	73	75	62	6d	69	74	0d	0a	41	63	63
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22	0100	4c	61	6e	67	75	61	67	65	3a	20	65	6e
Transmission Control Protocol, Src Port: 36044, Dst Port: 80, Seq: 1, Ack: 1, Len: 649	0110	0a	43	6f	6e	74	65	6e	74	2d	54	79	70

Fig10. Frame 83601, Ethernet II line: source MAC 00:17:f2:e2:c0:ce (Apple_e2:c0:ce).

The IP address identifies a network-layer endpoint, while the MAC address identifies the Ethernet interface observed as the source of the captured frames. In this case, the relevant traffic from 192.168.15.4 consistently carries source MAC 00:17:f2:e2:c0:ce, linking the DNS, HTTP, and cookie activity to the same observed interface. However, the network was open and shared, so neither the IP address nor the MAC address alone proves which person was using the device at the relevant time. The presence of three different User-Agent strings from the same MAC—Internet Explorer on Windows and Firefox/Safari on Mac OS X—suggests that the MAC may have been used with different browser or operating-system environments, or that the captured traffic includes more than one software context. A User-Agent is self-reported application data and should therefore be treated as supporting evidence rather than a reliable identifier of a person or physical device.

Q5. Who is associated with the device?

Item	Observation
Cookie pair	gmailchat=jcoachj@gmail.com/475090 in frame 83326, 2008-07-22 06:04:05.546193 UTC, 192.168.15.4 to 74.125.19.17, Host: mail.google.com
Persistence	TShark: 109 packets from the client MAC carry a cookie containing "jcoachj"; first frame 77528 (06:00:44.470524 UTC), last frame 84201 (06:04:55.380722 UTC); the POST (83601) falls inside this window
Competing identities	TShark pattern search for other email-shaped cookie values (.com, .net, .org, .edu) from this MAC returned 0 packets; limited to unencrypted HTTP cookies and those endings
False positives checked	Frame 18823 (www.amazon.com): "@" inside a random-looking token. Frame 59712 (adopt.specificklick.net): "@@" used as separators in an ad-tracking value. Neither is an email address

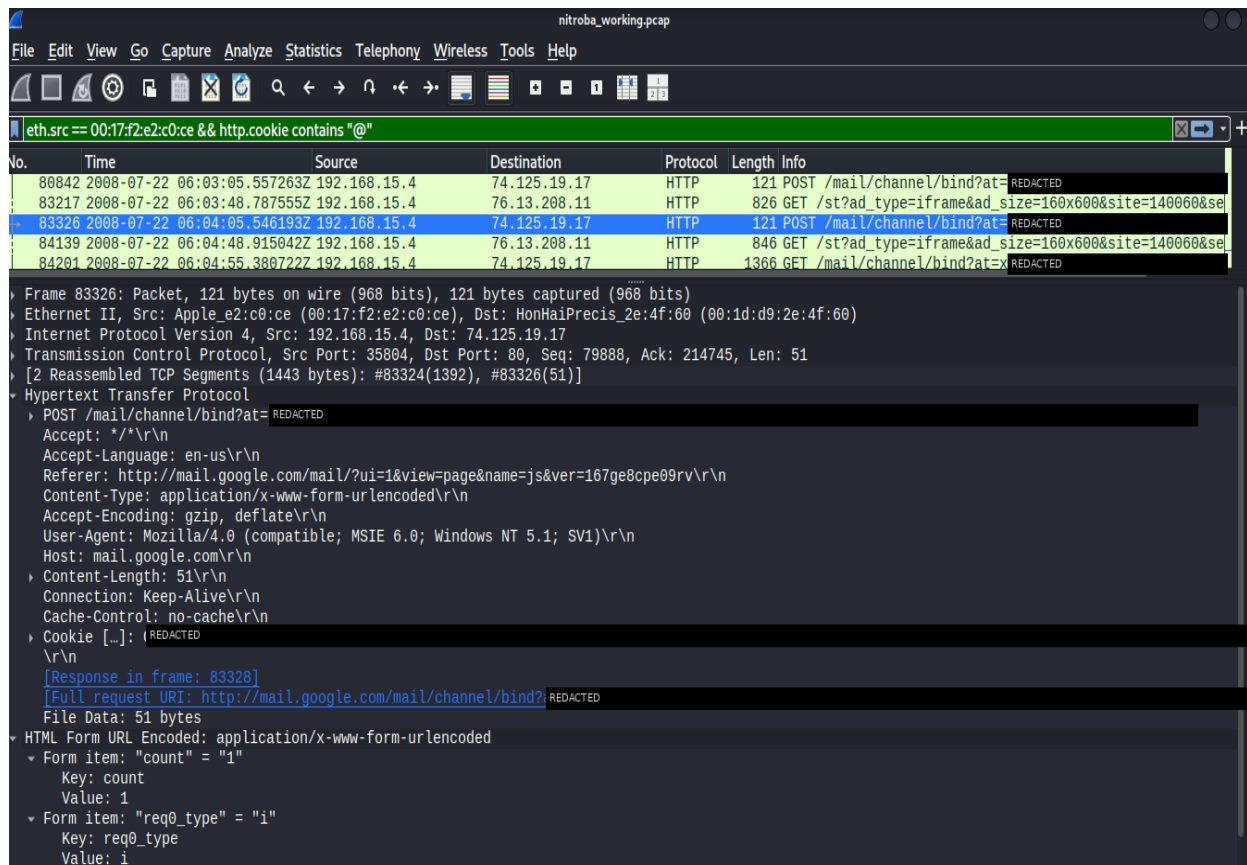


Fig11a. Filter eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@": frame 83326 to mail.google.com (session values redacted).

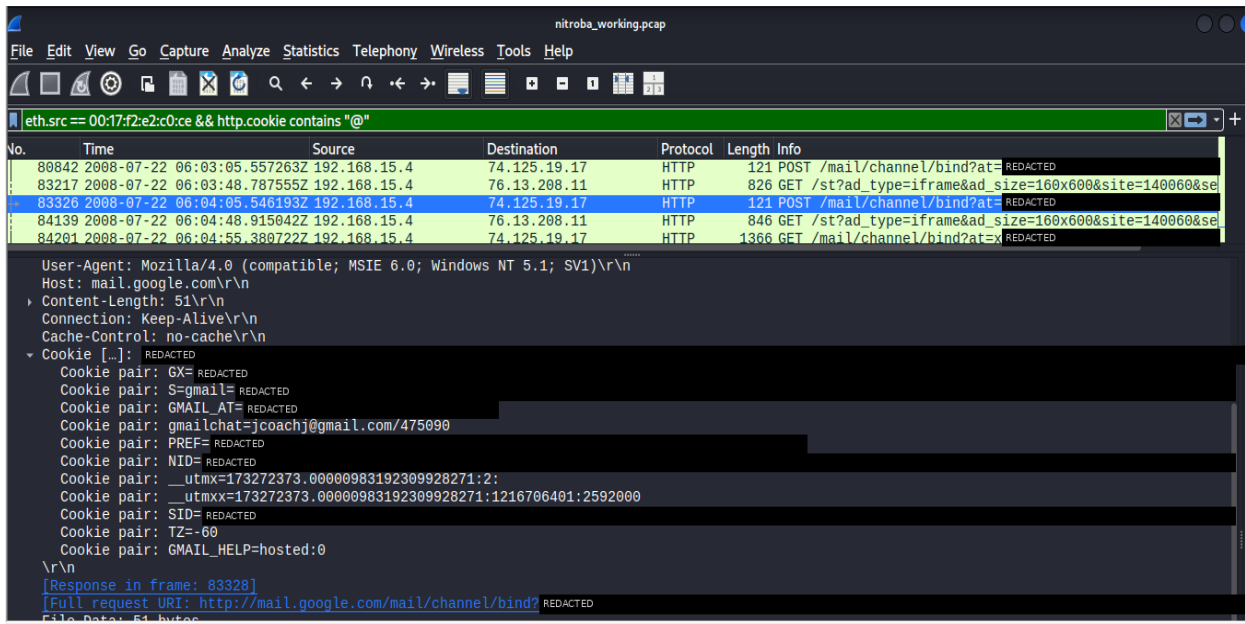


Fig11b. Frame 83326, cookie pairs: gmailchat=jcoachj@gmail.com/475090 (session values redacted).

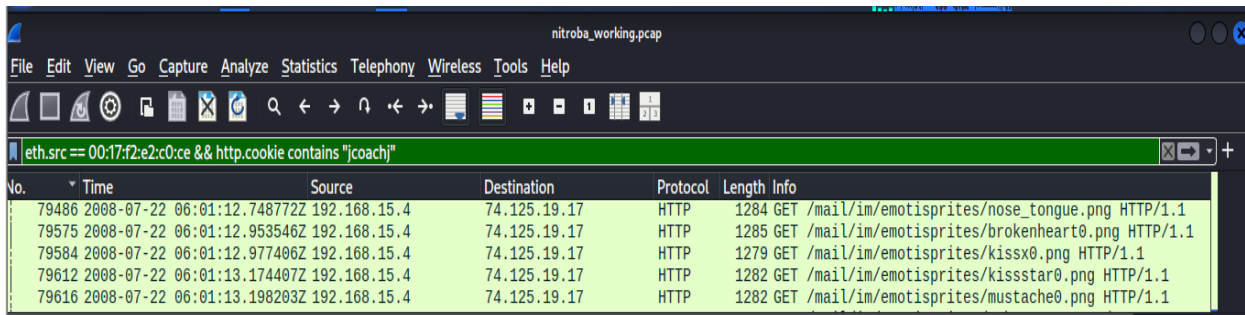


Fig11c. Wireshark list of packets from the client MAC whose cookie contains "jcoachj".



Fig12a. TShark counts: 109 packets with jcoachj; 35 packets with "@" excluding jcoachj and Amazon.

```
(kali@kali)-[~/DF204_case1]
$ tshark -r nitroba_working.pcap -Y 'eth.src == 00:17:f2:e2:c0:ce && http.cookie cont
hj_packets.txt
head -n 1 jcoachj_packets.txt
tail -n 1 jcoachj_packets.txt
wc -l jcoachj_packets.txt
77528      2008-07-22T06:00:44.470524000Z
84201      2008-07-22T06:04:55.380722000Z
109 jcoachj_packets.txt

(kali@kali)-[~/DF204_case1]
$ tshark -r nitroba_working.pcap -Y 'eth.src == 00:17:f2:e2:c0:ce && http.cookie matc
ns "jcoachj")' 2>/dev/null | wc -l
0

(kali@kali)-[~/DF204_case1]
$
```

Fig12b. First and last jcoachj packets (frames 77528 and 84201, UTC), total 109, and 0 other email-shaped cookie values.

nitroba_working.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@" && !(http.cookie contains "jcoachj")

No.	Time	Source	Destination	Protocol	Length	Info
18823	2008-07-22 04:28:53.812853Z	192.168.15.4	72.21.210.11	HTTP	763	GET / HTTP/1.1

Frame 18823: Packet, 763 bytes on wire (6104 bits), 763 bytes captured (6104 bits)

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 72.21.210.11

Transmission Control Protocol, Src Port: 32814, Dst Port: 80, Seq: 1, Ack: 1, Len: 765

Hypertext Transfer Protocol

GET / HTTP/1.1\r\n

User-Agent: Mozilla/5.0 (Macintosh; U; Intel Mac OS X 10_5_4; en-us) AppleWebKit/525.18 (KHTML, like Gecko) Version/3.1.2 Safari/525.20.1\r\n

Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,image/png,*/*;q=0.5\r\n

Accept-Language: en-us\r\n

Accept-Encoding: gzip, deflate\r\n

Cookie [...]: REDACTED

Cookie pair: ubid-main= REDACTED

Cookie pair: x-main=uiUngrTPA5eJ@UyKjyctbvG6GA48V1pp

Cookie pair: at-main= REDACTED

Cookie pair: session-id-time=12168828001

Cookie pair: session-id= REDACTED

Connection: keep-alive\r\n

Host: www.amazon.com\r\n

\r\n

[Response in frame: 18921]

Fig13a. False positive, frame 18823 (www.amazon.com): "@" inside a random-looking token (values redacted).

nitroba_working.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@" && !(http.cookie contains "jcoachj") && !(http.host contains "amazon")

No.	Time	Source	Destination	Protocol	Length	Info
59712	2008-07-22 05:48:24.973701Z	192.168.15.4	64.79.161.90	HTTP	376	GET /adopt.sm?l=653196995&sz=728x90&rnd=20358&r=j&xt=...

Frame 59712: Packet, 376 bytes on wire (3008 bits), 376 bytes captured (3008 bits)

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 64.79.161.90

Transmission Control Protocol, Src Port: 34910, Dst Port: 80, Seq: 1341, Ack: 1, Len: 318

[2 Reassembled TCP Segments (1658 bytes): #59710(1340), #59712(318)]

Hypertext Transfer Protocol

GET /adopt.sm?l=653196995&sz=728x90&rnd=20358&r=j&xt=&kw=&smuid=059-DyPp64 HTTP/1.1\r\n

Host: adopt.specificclick.net\r\n

User-Agent: Mozilla/5.0 (Macintosh; U; Intel Mac OS X; en-US; rv:1.8.1.16) Gecko/20080702 Firefox/2.0.0.16\r\n

Accept: */*\r\n

Accept-Language: en-us,en;q=0.7,he;q=0.3\r\n

Accept-Encoding: gzip,deflate\r\n

Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7\r\n

Keep-Alive: 300\r\n

Connection: keep-alive\r\n

Referer: http://www.orbitz.com/App/PerformDataPersistence?&retrieveParams=true&z=5015&r=2c\r\n

Cookie [...]: REDACTED

Cookie pair: smc= REDACTED

Cookie pair: dmp= REDACTED

Cookie pair: dmc= REDACTED

Cookie pair: dmk= REDACTED

Cookie pair: smx= REDACTED

Cookie pair: apfe=0A26mR

Cookie pair: smk=059-DyPp64

Cookie pair: CTCI=0018103Vb100iI10HIY10CTQ10191016s10000100W600000000

Cookie pair: UI=1BA1E001FC9BA496|06055.f.9.fh.xz.171@who@n.lmgv1vb@-1_9@f@ef|

Cookie pair: LO=00NuEH9x0M00C3000H0Wf0lwX1

Cookie pair: DMEXP=4

Cookie pair: DGI=-1|

\r\n

[Response in frame: 59716]

Fig13b. False positive, frame 59712 (adopt.specificclick.net): "@@" used as separators in an ad-tracking value (values redacted).

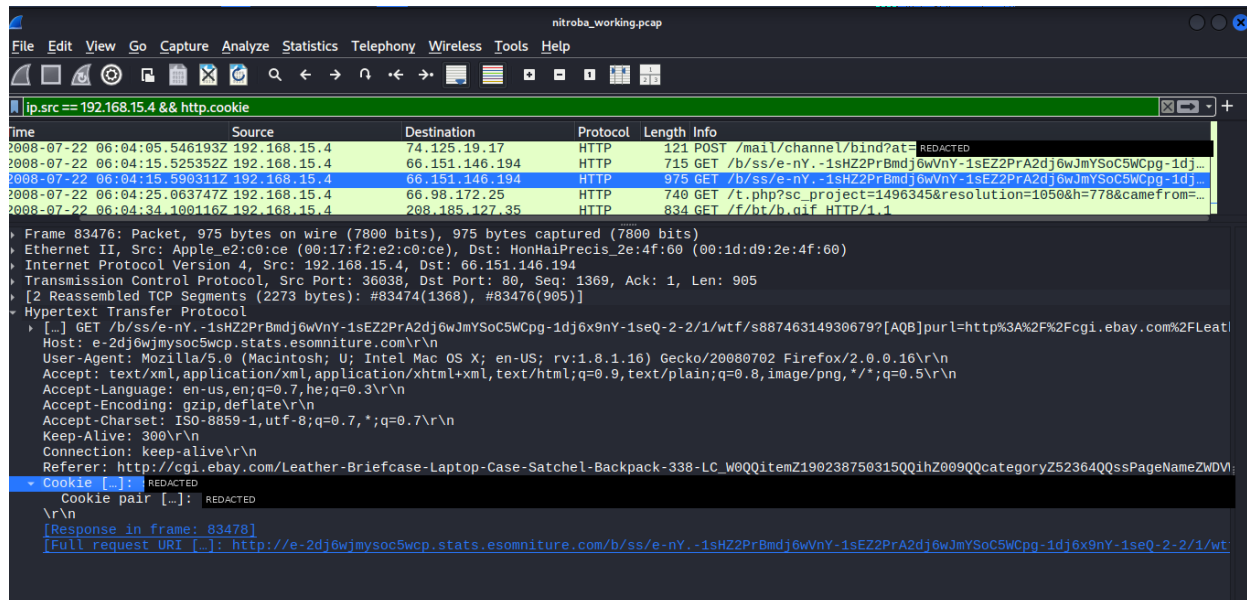


Fig13c. Frame 83476: Firefox 2.0.0.16 on Mac OS X from the same MAC (cookie values redacted).

The Gmail address jcoachj@gmail.com in the cookie indicates that an established Gmail session associated with that address was active in HTTP traffic from the same client MAC. Its evidential value is stronger than a single isolated packet because the value was observed in 109 packets over a continuous period from frame 77528 through frame 84201, and the harassment POST at frame 83601 falls within that period. This persistence strengthens the correlation between the Gmail session, the device, and the relevant activity. Nevertheless, the cookie does not prove who was physically at the keyboard: a session cookie can be reused by another person with access to the browser or device, and the capture does not provide direct evidence of the human user. The cookie is therefore strong corroborating evidence of account use, but not conclusive proof of personal identity.

Q6. Is that person on the Chem 109 roster?

Roster provided in the case slides (teacher Lily Tuckrige; students: Amy Smith, Burt Greedom, Tuck Gorge, Ava Book, Johnny Coach, Jeremy Ledvkin, Nancy Colburne, Tamara Perkins, Esther Pringle, Asar Misrad, Jenny Kant). The roster lists names only, not email addresses (Figure 14).

2. Chem 109 - Class roster

Teacher: Lily Tuckrige

Students:

Amy Smith
Burt Greedom
Tuck Gorge
Ava Book
Johnny Coach
Jeremy Ledvkin
Nancy Colburne
Tamara Perkins
Esther Pringle
Asar Misrad
Jenny Kant

Fig14. Chem 109 class roster from the case slides.

The address jcoachj@gmail.com most closely resembles the Chem 109 roster name Johnny Coach. The substring “jcoach” corresponds closely to the initial “J” and surname “Coach”, while the final “j” in the address does not correspond directly to the roster name. I also considered the other names on the roster, but none provides a similarly close textual relationship to the address. This is an inference rather than a verified identification because the roster provides names only and does not establish that Johnny Coach owned or controlled the Gmail account. The roster comparison therefore provides moderate corroboration when considered together with the persistent Gmail cookie and network evidence, but it should not be stated as proof that Johnny Coach was the person at the keyboard.

Q7. When did the activity occur? (all times UTC, from the PCAP)

UTC time	Frame	Event	Evidence
2008-07-22 01:51:07.095278	1	First packet in the capture	E02
2008-07-22 06:00:44.470524	77528	First packet carrying the jcoachj cookie (first observed)	E11
2008-07-22 06:03:43.640466	82912	DNS query for www.willselfdestruct.com	E03
2008-07-22 06:03:43.720123	82923	DNS response: 69.25.94.22	E03
2008-07-22 06:03:43.825871	82936	HTTP GET /secure/submit	E04
2008-07-22 06:03:44.210946	82998	HTTP 200 OK from the server (385.075 ms after the request)	E05
2008-07-22 06:04:05.546193	83326	Gmail request carrying the jcoachj cookie	E10
2008-07-22 06:04:24.311700	83601	HTTP POST containing the message	E06
Not independently recorded in the supplied evidence; the 302 response is confirmed in tcp.stream 1707 / Fig08a.	Server response to the POST: HTTP/1.1 302 Moved Temporarily, Location /secure/success (tcp.stream 1707). The exact frame number is not stated elsewhere in the supplied report.	Server response to the POST: 302 Moved Temporarily (tcp.stream 1707)	E07
2008-07-22 06:04:55.380722	84201	Last packet carrying the jcoachj cookie (last observed)	E11

Time zone: Wireshark was set to UTC (trailing Z). Values from the Capture File Properties window were converted from EDT by adding 4 hours. The server's own Date headers (for example 07:24:05 GMT on the 200 OK and 07:24:45 GMT on the 302) run about 1 hour 20 minutes ahead of the capture clock, with the same 40-second spacing as the capture; the timeline uses the capture timestamps.

Q8. What conclusion can you defend?

The evidence shows that client 192.168.15.4 accessed willselfdestruct.com and subsequently submitted the harassment message, and that the same observed MAC address carried a persistent Gmail cookie containing jcoachj@gmail.com during the relevant period. This suggests that the device and network session were associated with the account represented by jcoachj@gmail.com, which in turn resembles the Chem 109 roster name Johnny Coach. However, the evidence does not conclusively identify the person who operated the device because the wireless network was open and shared, the cookie identifies an account session rather than a human, and device or browser identifiers can be altered or shared. Confidence: Moderate, because several independent network and application indicators agree, but the final person-level attribution remains inferential.

5. Attribution Assessment

5.1 Observed facts

Observed fact	Frames	Evidence
The client at 192.168.15.4 looked up and then requested www.willselfdestruct.com (69.25.94.22)	82912, 82923, 82936	E03, E04
The same client sent an HTTP POST with the harassing message to the service at 06:04:24.311700 UTC	83601	E06, E07
The POST content matches the message shown in the teacher's second email	83601	E08
The Ethernet source of the client frames is 00:17:f2:e2:c0:ce	82936, 83601	E09
A Gmail cookie containing jcoachj@gmail.com was sent from the same MAC in 109 packets, including shortly before and after the POST	77528 to 84201	E10, E11
No other email-shaped cookie value was found from this MAC in the searched HTTP cookies	-	E12

5.2 Reasonable inferences

- 1. Inference:** The device at 192.168.15.4 was responsible for submitting the second harassment message because it resolved and contacted willselfdestruct.com and sent the matching POST in frame 83601; the same client MAC appears on the relevant traffic.
- 2. Inference:** The device was being used in a browser session associated with jcoachj@gmail.com at the time of the harassment activity because the Gmail cookie was observed from the same MAC in 109 packets, with the POST occurring within the observed cookie period.
- 3. Inference:** The account may be associated with Johnny Coach, a Chem 109 student, because jcoachj@gmail.com closely resembles that roster name. This is the weakest attribution step because the roster contains names only and does not independently verify ownership or use of the account.

5.3 Limitations and alternative explanations

The principal limitation is that the traffic was captured on an open, shared wireless network. The evidence therefore establishes a device and network session more strongly than it establishes a particular person. A second limitation is that a Gmail cookie identifies a browser session associated with an account, but another person with access to the device or session could potentially have generated traffic using it. MAC addresses and User-Agent strings are useful correlation indicators but are not immutable proof of identity and can be changed, spoofed, or shared. The search for competing email-shaped cookies was also limited to unencrypted HTTP cookies and the specified domain endings, so it cannot exclude identities communicated through encrypted traffic or artifacts outside those search conditions. Finally, the capture represents observed network traffic; it does not by itself establish who physically controlled the device at the time of the request. These limitations are why the attribution is assessed as moderate rather than high confidence.

Appendix A - Evidence Log

ID	Packet / item	Finding	Why it matters	Screenshot
E01	nitroba.pcap and working copy	56,180,821 bytes; SHA-256 2b77a9ea...c67553ec2fb for both; cmp IDENTICAL; original read-only	Shows that the evidence was preserved and that the analysed working copy is identical to the original, supporting integrity of the examination.	Fig01

ID	Packet / item	Finding	Why it matters	Screenshot
E02	Capture File Properties	Same SHA-256; 2008-07-22 01:51:07 to 06:13:47 UTC; elapsed 04:22:39	Confirms the capture identity, hash value, and UTC examination timeline used throughout the report.	Fig02
E03	Frames 82912, 82923	DNS query and response: www.willselfdestruct.com -> 69.25.94.22 via resolver 192.168.1.254	Links the hostname to the service IP and identifies the client involved in the DNS resolution.	Fig03, Fig04
E04	Frame 82936	GET /secure/submit, Host www.willselfdestruct.com, 192.168.15.4 to 69.25.94.22, 06:03:43.825871 UTC	Provides application-layer evidence that the client requested willselfdestruct.com and identifies the exact URI and time.	Fig05
E05	Frame 82998	HTTP/1.1 200 OK from 69.25.94.22 at 06:03:44.210946 UTC; Set-Cookie JSESSIONID	Confirms a successful HTTP response from the identified service and establishes the related web session.	Fig06
E06	Frame 83601	POST /secure/submit; to=lilytuckrige@yahoo.com, from empty, subject "you can't find us", message three lines; 06:04:24.311700 UTC	Directly contains the recipient, subject and message submitted to the willselfdestruct.com service, making it the key message-content evidence.	Fig07

ID	Packet / item	Finding	Why it matters	Screenshot
E07	tcp.stream 1707	Request (Content-Length 188) and 302 Moved Temporarily to /secure/success; the supplied report does not independently record the response frame number.	Reconstructs the complete transaction and confirms that the server returned a 302 redirect after receiving the POST. The supplied report does not independently record the response frame number.	Fig08
E08	Teacher's second email vs frame 83601	To, Subject and message match; sender shown as [Undisclosed Sender] = empty from field	Corroborates the HTTP request length and supports completeness of the reconstructed POST body.	Fig09
E09	Frames 82936, 83601	Source MAC 00:17:f2:e2:c0:ce (Apple_e2:c0:ce); destination 00:1d:d9:2e:4f:60	Provides an independent visual comparison showing that the captured POST content matches the message displayed to the teacher.	Fig10
E10	Frame 83326	Cookie pair gmailchat=jcoachj@gmail.com/475090 to mail.google.com at 06:04:05.546193 UTC	Links the relevant network traffic to a consistent Ethernet source interface and supports device-level correlation.	Fig11
E11	TShark count	109 packets with jcoachj; first frame 77528 (06:00:44.470524 UTC), last 84201 (06:04:55.380722 UTC)	Shows that the same client MAC was using a Gmail session associated with jcoachj@gmail.com shortly before the harassment POST.	Fig12

ID	Packet / item	Finding	Why it matters	Screenshot
E12	TShark pattern search	0 other email-shaped cookie values from this MAC	Demonstrates that the jcoachj cookie value persisted across 109 packets and that the POST occurred within that observed period.	Fig12
E13	Frames 18823, 59712, 83476	False positives (Amazon, ad network) and three browser identities from one MAC	Shows that apparent alternative email-shaped matches were not treated as identities without inspection, reducing the risk of false attribution.	Fig13
E14	Frames 82998, 302 response	Server Date header about 1 h 20 min ahead of capture clock; consistent offset		Fig06, Fig08

Appendix B - Figure Index

All screenshots show the active filter, the packet list, the relevant packet detail and the time. Long session tokens (GX, SID, NID, PREF, at=, cookie values) were redacted in Fig11a, Fig11b, Fig13a, Fig13b and Fig13c; the identifying cookie pair in Fig11b is left visible. Image files are in the screenshots/ folder.

Figure	Shows	Used in	Evidence
Fig01a	Download of nitroba.pcap with wget (56,180,821 bytes) and file listing.	Section 2	E01
Fig01b	SHA-256 of the original, creation of the working copy, SHA-256 of the copy, and chmod 444 on the original.	Section 2	E01
Fig01c	cmp reports IDENTICAL; the original is read-only (-r--r--r--); date shows EDT and UTC; Wireshark opened on the working copy.	Section 2	E01
Fig02a	Statistics > Capture File Properties: SHA-256, SHA-1, pcap format, Ethernet encapsulation, first and last packet (VM local time) and elapsed time.	Section 2	E02
Fig02b	Packet list with the time display set to UTC (trailing Z); first packet 2008-07-22 01:51:07.095278Z.	Section 2	E02
Fig03a	Filter frame contains "willselfdestruct": first matching packets from 192.168.15.4.	Q2	E03, E04

Figure	Shows	Used in	Evidence
Fig03b	Same filter with the Protocol and Info columns: DNS query 82912, DNS response 82923, HTTP GET /secure/submit 82936.	Q2	E03, E04
Fig04	DNS response, frame 82923: Answers show the CNAME and the A record 69.25.94.22.	Q2	E03
Fig05	HTTP GET, frame 82936: Host www.willselfdestruct.com and the Full request URI.	Q2	E04
Fig06	HTTP/1.1 200 OK, frame 82998, filter ip.addr == 69.25.94.22 && http, with Request in frame 82936.	Q2	E05, E14
Fig07a	Filter ip.addr == 69.25.94.22 && http.request.method == "POST": one packet, frame 83601.	Q3	E06
Fig07b	Frame 83601, HTML Form URL Encoded: to, from, subject, message, type, ttl, submit.x and submit.y.	Q3	E06
Fig08a	Follow TCP Stream (tcp.stream eq 1707): POST request and 302 Moved Temporarily response.	Q3	E07, E14
Fig08b	Body length check: the body is 188 bytes, matching Content-Length 188.	Q3	E07, E14
Fig09	The teacher's second email as displayed by willselfdestruct.com (From: [Undisclosed Sender]).	Q3	E08
Fig10	Frame 83601, Ethernet II line: source MAC 00:17:f2:e2:c0:ce (Apple_e2:c0:ce).	Q4	E09
Fig11a	Filter eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@": frame 83326 to mail.google.com (session values redacted).	Q5	E10, E11
Fig11b	Frame 83326, cookie pairs: gmailchat=jcoachj@gmail.com/475090 (session values redacted).	Q5	E10, E11
Fig11c	Wireshark list of packets from the client MAC whose cookie contains "jcoachj".	Q5	E10, E11
Fig12a	TShark counts: 109 packets with jcoachj; 35 packets with "@" excluding jcoachj and Amazon.	Q5	E11, E12
Fig12b	First and last jcoachj packets (frames 77528 and 84201, UTC), total 109, and 0 other email-shaped cookie values.	Q5	E11, E12
Fig13a	False positive, frame 18823 (www.amazon.com): "@" inside a random-looking token (values redacted).	Q5	E13
Fig13b	False positive, frame 59712 (adopt.specificclick.net): "@@" used as separators in an ad-tracking value (values redacted).	Q5	E13

Figure	Shows	Used in	Evidence
Fig13c	Frame 83476: Firefox 2.0.0.16 on Mac OS X from the same MAC (cookie values redacted).	Q5	E13
Fig14	Chem 109 class roster from the case slides.	Q6	Q6

Appendix C - Repository and File Inventory

File	Contents
evidence/nitroba.pcap	Original evidence (kept locally, read-only; not uploaded)
working/nitroba_working.pcap	Working copy used for all analysis (not uploaded)
reports/evidence_hashes.txt	SHA-256 logs for original and working evidence
reports/http_requests.txt	HTTP GET/POST log from TShark
reports/tcp_streams.txt	Reconstructed TCP streams for the GET and POST sessions
reports/endpoint_mac_mapping.tsv	MAC to IP address mapping
reports/cookie_artifacts.txt	Cookie and browser identity indicators
screenshots/	Numbered screenshots Fig01 to Fig14
scripts/generate_report_files.sh	Regenerates the reports/ files from the working copy

Student Declaration

I confirm that I completed this case study individually, worked only on a copy of the supplied evidence, did not contact any host, account or person referenced in the evidence, redacted unrelated personal data, and that the interpretation, wording and conclusions in this report are my own.